Appendix: Theoretical Analysis of the Augmented Synthetic Historical Control Estimator (ASHC)

Setup and Assumptions

For the purposes of self-containment, let $\mathcal{T} = \{1, 2, \ldots, T\}$ index time at a monthly frequency. Define the pre-treatment period as $\mathcal{T}_1 := \{t \in \mathcal{T} : t \leq T_0\}$ and the post-treatment period as $\mathcal{T}_2 := \{t \in \mathcal{T} : t > T_0\}$. The number of post-treatment periods is $n = T - T_0$. Let $m \in \mathbb{N}$ denote the evaluation window length. Define the evaluation period as the final m months of the pre-treatment period $\mathcal{T}_{\text{eval}} := \{T_0 - m + 1, \ldots, T_0\} \subset \mathcal{T}_1$.

Let $\mathbf{y} = (y_1, y_2, \ldots, y_T)^\top \in \mathbb{R}^T$ denote the observed outcome vector for the treated unit. Define the pre-treatment outcome vector $\mathbf{y}_{\text{pre}} := \mathbf{y}_{\mathcal{T}_1} \in \mathbb{R}^{T_0}$, the evaluation subvector $\mathbf{y}_{\text{eval}} := \mathbf{y}_{\mathcal{T}_{\text{eval}}} \in \mathbb{R}^m$, and the post-treatment outcome vector $\mathbf{y}_{\text{post}} := \mathbf{y}_{\mathcal{T}_2} \in \mathbb{R}^n$.

The evaluation subvector $\mathbf{y}_{\text{eval}}$ is the target pattern to be reconstructed using convex combinations of earlier segments from the pre-treatment period.

SHC by Chen et al. [2024] constructs

$$N = T_0 - m - n + 1$$

overlapping temporal segments of length m. Each segment $\widetilde{\mathbf{y}}_{[i,i+m-1]}$ is a contiguous subvector of the smoothed pre-treatment series. These comprise the donor matrix

$$\widetilde{\mathbf{Y}}_{\text{pre}} := \begin{bmatrix} \widetilde{\mathbf{y}}_{[1,m]} & \widetilde{\mathbf{y}}_{[2,m+1]} & \cdots & \widetilde{\mathbf{y}}_{[N,N+m-1]} \end{bmatrix} \in \mathbb{R}^{m \times N}.$$

Example 0.1. *Suppose we have $T_0 = 60$ months of pre-treatment data and we wish to construct donor segments of length $m = 12$ months to predict a post-treatment horizon of $n = 6$ months. The number of eligible segments is:*

$$N = 60 - 12 - 6 + 1 = 43.$$

Each donor segment is a vector of 12 consecutive smoothed pre-treatment outcomes.

In the third step, we solve the following program to obtain the optimal weights:

$$\begin{aligned} \mathbf{w}^* = \underset{\mathbf{w} \in \mathbb{R}^N}{\operatorname{argmin}} \quad & \left\| \widetilde{\mathbf{y}}_{\text{eval}} - \widetilde{\mathbf{Y}}_{\text{pre}} \mathbf{w} \right\|_2^2 + \varsigma \left\| \mathbf{C}_0^\top \mathbf{w} \right\|_2^2 \\ \text{subject to} \quad & \mathbf{w} \geq \mathbf{0}, \quad \mathbf{1}^\top \mathbf{w} = 1. \end{aligned}$$

Let $\mathbf{w}^{\text{SHC}}$ denote the SHC solution, and $\mathbf{w}^{\text{ASHC}}$ the ASHC solution:

$$\mathbf{w}^{\text{ASHC}} = \arg\min_{\mathbf{w} \in \mathbb{R}^N} \left\{ \frac{1}{2\lambda} \|\widetilde{\mathbf{y}} - \widetilde{\mathbf{Y}} \mathbf{w}\|^2 + \|\mathbf{w} - \mathbf{w}^{\text{SHC}}\|^2 + \varsigma \|\mathbf{C}_0^\top \mathbf{w}\|_2^2 \right\} \quad \text{s.t.} \ \sum_{i=1}^N w_i = 1,$$

where $\lambda > 0$ is the regularization parameter, $\varsigma > 0$ is the penalty parameter for the low-variance directions, and $\mathbf{C}_0 \in \mathbb{R}^{N \times k}$ contains the eigenvectors of the Gram matrix $\mathbf{G} = \widetilde{\mathbf{Y}}^\top \widetilde{\mathbf{Y}}$ corresponding to eigenvalues less than a threshold $\tau > 0$.

Assumptions

Assumption 0.1 (Linearity with noise)**.** *There exists* $\mathbf{w}^* \in \mathbb{R}^N$ *such that*

$$\widetilde{\mathbf{y}} = \widetilde{\mathbf{Y}}\mathbf{w}^* + \boldsymbol{\eta},$$

where $\boldsymbol{\eta} \in \mathbb{R}^m$ *is a mean-zero noise vector.*

Assumption 0.2 (Bounded noise)**.** $\|\boldsymbol{\eta}\| \leq \delta$ *for some* $\delta > 0$*.*

Assumption 0.3 (Smoothness of latent trend)**.** *Let* $\ell(t)$ *denote the latent smooth component of the treated potential outcome. Suppose* $\ell(t)$ *is* $H+1$ *times differentiable with* $|\ell^{(H+1)}(t)| < \epsilon$ *for all* t *and some* $H \geq 3$*.*

Assumption 0.4 (Operator norm bound)**.** $\|\widetilde{\mathbf{Y}}\| \leq \gamma$*.*

Assumption 0.5 (Simplex feasibility)**.** $\mathbf{w}^{SHC} \in \Delta^{N-1} := \{\mathbf{w} \in \mathbb{R}^N : \sum w_i = 1,\ w_i \geq 0\}$*.*

Lemma 0.1 (Deviation from SHC)**.** *Let* $\Delta := \mathbf{w}^{ASHC} - \mathbf{w}^{SHC}$*. Then*

$$\|\Delta\| \ \leq \ \frac{\gamma}{2\lambda + 2\varsigma\|\mathbf{C}_0\|^2}\, \epsilon_{SHC}.$$

where $\epsilon_{SHC} := \|\widetilde{\mathbf{y}} - \widetilde{\mathbf{Y}}\mathbf{w}^{SHC}\|$ *and* $\|\mathbf{C}_0\|^2$ *is the operator norm of* $\mathbf{C}_0^\top \mathbf{C}_0$*.*

Proof. To bound in sample deviation, we can begin from the ASHC optimization problem:

$$J(\mathbf{w}) = \frac{1}{2\lambda}\|\widetilde{\mathbf{y}} - \widetilde{\mathbf{Y}}\mathbf{w}\|^2 + \|\mathbf{w} - \mathbf{w}^{\mathrm{SHC}}\|^2 + \varsigma\|\mathbf{C}_0^\top \mathbf{w}\|_2^2 \quad \text{s.t. } \mathbf{1}^\top \mathbf{w} = 1.$$

The three components have clear roles: the first encourages close fit to the evaluation period outcomes, the second penalizes deviation from the SHC weights, and the third stabilizes the solution in low-variance directions.

If we minimized $J(\mathbf{w})$ without the affine constraint $\mathbf{1}^\top \mathbf{w} = 1$, the resulting $\mathbf{w}$ might not be feasible. To enforce this constraint, we introduce the Lagrangian

$$\mathcal{L}(\mathbf{w}, \mu) = J(\mathbf{w}) + \mu(\mathbf{1}^\top \mathbf{w} - 1),$$

where $\mu \in \mathbb{R}$ is a Lagrange multiplier. This term acts as a corrective force, ensuring feasibility without changing the curvature of $J(\mathbf{w})$ that governs the size of Δ. We now may compute the gradient of $\mathcal{L}$.

First, for the fit term, define $r(\mathbf{w}) = \widetilde{\mathbf{y}} - \widetilde{\mathbf{Y}}\mathbf{w}$. Using the *chain rule for vector functions*,

$$\nabla_{\mathbf{w}} \frac{1}{2\lambda}\|r(\mathbf{w})\|^2 = \frac{1}{\lambda}\widetilde{\mathbf{Y}}^\top(\widetilde{\mathbf{Y}}\mathbf{w} - \widetilde{\mathbf{y}}).$$

Second, for the deviation penalty, recall the rule for the gradient of a quadratic form, $\nabla_{\mathbf{w}}(\mathbf{w}^\top A\mathbf{w}) = (A + A^\top)\mathbf{w}$. With $A = I$, we obtain

$$\nabla_{\mathbf{w}}\|\mathbf{w} - \mathbf{w}^{\mathrm{SHC}}\|^2 = 2(\mathbf{w} - \mathbf{w}^{\mathrm{SHC}}).$$

Third, for the low-variance penalty,

$$\varsigma\|\mathbf{C}_0^\top \mathbf{w}\|^2 = \varsigma \mathbf{w}^\top(\mathbf{C}_0\mathbf{C}_0^\top)\mathbf{w}.$$

Since $\mathbf{C}_0\mathbf{C}_0^\top$ is symmetric, applying the quadratic form rule yields

$$\nabla_{\mathbf{w}}\varsigma\|\mathbf{C}_0^\top \mathbf{w}\|^2 = 2\varsigma\mathbf{C}_0\mathbf{C}_0^\top \mathbf{w}.$$

Finally, differentiating the affine constraint term $\mu(\mathbf{1}^\top \mathbf{w} - 1)$, linearity of differentiation gives

$$\nabla_{\mathbf{w}}[\mu(\mathbf{1}^\top \mathbf{w} - 1)] = \mu\mathbf{1}.$$

Collecting terms, the stationarity condition $\nabla_{\mathbf{w}}\mathcal{L}(\mathbf{w},\mu)=0$ is

$$\left(\frac{1}{\lambda}\widetilde{\mathbf{Y}}^\top\widetilde{\mathbf{Y}}+2\mathbf{I}+2\varsigma\mathbf{C}_0\mathbf{C}_0^\top\right)\mathbf{w}=\frac{1}{\lambda}\widetilde{\mathbf{Y}}^\top\widetilde{\mathbf{y}}+2\mathbf{w}^{\text{SHC}}-\mu\mathbf{1}.$$

Now with our FOC, we must express our deviation term

$$\Delta:=\mathbf{w}^{\text{ASHC}}-\mathbf{w}^{\text{SHC}}$$

in terms of known quantities. To achieve this, we first package the curvature terms of the Lagrangian into a single matrix:

$$M:=\tfrac{1}{\lambda}\widetilde{\mathbf{Y}}^\top\widetilde{\mathbf{Y}}+2\mathbf{I}+2\varsigma\mathbf{C}_0\mathbf{C}_0^\top.$$

From the first-order condition for $\mathbf{w}^{\text{ASHC}}$, we obtained:

$$M\mathbf{w}^{\text{ASHC}}=\tfrac{1}{\lambda}\widetilde{\mathbf{Y}}^\top\widetilde{\mathbf{y}}+2\mathbf{w}^{\text{SHC}}-\mu\mathbf{1}. \tag{FOC-ASHC}$$

We now evaluate the same structure at $\mathbf{w}^{\text{SHC}}$. Note that $\mathbf{w}^{\text{SHC}}$ need not satisfy the ASHC first-order condition, but writing the analogous expression will allow us to cancel common terms. Plugging $\mathbf{w}^{\text{SHC}}$ into the left-hand side yields:

$$M\mathbf{w}^{\text{SHC}}=\tfrac{1}{\lambda}\widetilde{\mathbf{Y}}^\top\widetilde{\mathbf{Y}}\mathbf{w}^{\text{SHC}}+2\mathbf{w}^{\text{SHC}}+2\varsigma\mathbf{C}_0\mathbf{C}_0^\top\mathbf{w}^{\text{SHC}}. \tag{Baseline}$$

Subtracting (Baseline) from (FOC-ASHC), we obtain:

$$M(\mathbf{w}^{\text{ASHC}}-\mathbf{w}^{\text{SHC}})=\tfrac{1}{\lambda}\widetilde{\mathbf{Y}}^\top(\widetilde{\mathbf{y}}-\widetilde{\mathbf{Y}}\mathbf{w}^{\text{SHC}})-\mu\mathbf{1}.$$

That is,

$$M\Delta=\tfrac{1}{\lambda}\widetilde{\mathbf{Y}}^\top\big(\widetilde{\mathbf{y}}-\widetilde{\mathbf{Y}}\mathbf{w}^{\text{SHC}}\big)-\mu\mathbf{1}. \tag{Deviation Equation}$$

This subtraction step isolates the deviation Δ, which is the central quantity we need to bound and expresses Δ entirely in terms of the SHC residual $r:=\widetilde{\mathbf{y}}-\widetilde{\mathbf{Y}}\mathbf{w}^{\text{SHC}}$ and the Lagrange multiplier μ, eliminating any direct dependence on the unknown $\mathbf{w}^{\text{ASHC}}$.

The affine constraint $\mathbf{1}^\top\mathbf{w}^{\text{ASHC}}=1$ and $\mathbf{1}^\top\mathbf{w}^{\text{SHC}}=1$ together imply

$$\mathbf{1}^\top\Delta=\mathbf{1}^\top(\mathbf{w}^{\text{ASHC}}-\mathbf{w}^{\text{SHC}})=0.$$

This means Δ lies in the hyperplane orthogonal to the all-ones vector $\mathbf{1}$. Thus any component of the right-hand side parallel to $\mathbf{1}$ (such as $-\mu\mathbf{1}$) does not affect Δ within its feasible subspace.

To make this explicit, take inner products of both sides of (Deviation Equation) with Δ:

$$\Delta^\top M\Delta=\tfrac{1}{\lambda}\Delta^\top\widetilde{\mathbf{Y}}^\top\big(\widetilde{\mathbf{y}}-\widetilde{\mathbf{Y}}\mathbf{w}^{\text{SHC}}\big)-\mu\underbrace{\Delta^\top\mathbf{1}}_{=0}.$$

Hence the $\mu\mathbf{1}$ term vanishes:

$$\Delta^\top M\Delta=\tfrac{1}{\lambda}\Delta^\top\widetilde{\mathbf{Y}}^\top\big(\widetilde{\mathbf{y}}-\widetilde{\mathbf{Y}}\mathbf{w}^{\text{SHC}}\big).$$

The Lagrange multiplier μ acts like a force that keeps $\mathbf{w}^{\text{ASHC}}$ on the feasible hyperplane $\{\mathbf{w}:\mathbf{1}^\top\mathbf{w}=1\}$. But since Δ itself lies entirely within this hyperplane, μ does not change Δ's magnitude. It enforces feasibility but does not distort the conditioning of M, which governs the bound on $\|\Delta\|$.

Let $r:=\widetilde{\mathbf{y}}-\widetilde{\mathbf{Y}}\mathbf{w}^{\text{SHC}}$ denote the residual from the SHC fit, with norm $\|r\|=\epsilon_{\text{SHC}}$. From the previous step, we now have

$$\Delta^\top M\Delta=\tfrac{1}{\lambda}\Delta^\top\widetilde{\mathbf{Y}}^\top r.$$

Applying the Cauchy–Schwarz inequality, we obtain

$$\Delta^\top M\Delta \;\le\; \tfrac{1}{\lambda}\|\Delta\|\,\|\widetilde{\mathbf{Y}}^\top r\|.$$

By Assumption (A4), the operator norm of $\widetilde{\mathbf{Y}}$ is bounded:

$$\|\widetilde{\mathbf{Y}}^\top r\| \;\le\; \|\widetilde{\mathbf{Y}}\|\,\|r\| \;\le\; \gamma\,\epsilon_{\mathrm{SHC}}.$$

Hence,

$$\Delta^\top M\Delta \;\le\; \frac{\gamma}{\lambda}\,\|\Delta\|\,\epsilon_{\mathrm{SHC}}.$$

At the same time, since M is positive definite, we can bound the left-hand side below using its smallest eigenvalue. Let $\lambda_{\min}(M)$ denote the minimum eigenvalue of M. Then

$$\Delta^\top M\Delta \;\ge\; \lambda_{\min}(M)\,\|\Delta\|^2.$$

Combining the two bounds gives

$$\lambda_{\min}(M)\,\|\Delta\|^2 \;\le\; \frac{\gamma}{\lambda}\,\|\Delta\|\,\epsilon_{\mathrm{SHC}}.$$

Dividing through by $\|\Delta\| > 0$ (the case $\Delta = 0$ being trivial) yields

$$\|\Delta\| \;\le\; \frac{\gamma}{\lambda\,\lambda_{\min}(M)}\,\epsilon_{\mathrm{SHC}}.$$

Finally, recall that M was defined as

$$M = \tfrac{1}{\lambda}\widetilde{\mathbf{Y}}^\top\widetilde{\mathbf{Y}} + 2\mathbf{I} + 2\varsigma\mathbf{C}_0\mathbf{C}_0^\top.$$

The first two terms ensure $\lambda_{\min}(M) \ge 2$, and the stabilizing penalty contributes at least $2\varsigma\,\lambda_{\min}(\mathbf{C}_0\mathbf{C}_0^\top)$. For simplicity, we use the operator norm bound $\|\mathbf{C}_0\|^2 = \lambda_{\max}(\mathbf{C}_0^\top\mathbf{C}_0)$, so that

$$\lambda_{\min}(M) \;\ge\; 2 + 2\varsigma\|\mathbf{C}_0\|^2.$$

Substituting this bound, we obtain our final ASHC bound as:

$$\|\Delta\| \;\le\; \frac{\gamma}{2\lambda + 2\varsigma\|\mathbf{C}_0\|^2}\,\epsilon_{\mathrm{SHC}}.$$

This shows that the deviation of ASHC weights from the SHC baseline is tightly controlled by the residual error of SHC, scaled down by the effective regularization strength coming from both λ and the stabilizing penalty ς. □

Lemma 0.2. *Under (A4), the error from deviating off SHC weights satisfies*

$$\|\widetilde{\mathbf{Y}}\Delta\| \le \gamma\|\Delta\|.$$

Proof. By definition, $\Delta = \mathbf{w}^{\mathrm{ASHC}} - \mathbf{w}^{\mathrm{SHC}} \in \mathbb{R}^N$. From Assumption (A4), the operator (spectral) norm of the donor matrix satisfies

$$\|\widetilde{\mathbf{Y}}\| \;\le\; \gamma,$$

where the operator norm is

$$\|\widetilde{\mathbf{Y}}\| \;=\; \sup_{\mathbf{v}\neq 0}\frac{\|\widetilde{\mathbf{Y}}\mathbf{v}\|}{\|\mathbf{v}\|}.$$

Applying this inequality to $\mathbf{v} = \Delta$ gives

$$\|\widetilde{\mathbf{Y}}\Delta\| \;\le\; \|\widetilde{\mathbf{Y}}\|\,\|\Delta\| \;\le\; \gamma\|\Delta\|.$$

This establishes the claim. □

Remark 0.1. *Lemma 0.2 ensures that deviations of the ASHC weights from the SHC weights, measured in ℓ_2 norm, translate into controlled deviations in the predicted outcomes. The operator norm bound (A4) prevents $\widetilde{\mathbf{Y}}$ from amplifying small changes in the weights disproportionately.*

Lemma 0.3. *Under (A3), for horizon $k > 0$,*

$$|\ell(T_0 + k) - \ell^{SHC}(T_0 + k)| \;\leq\; b_\epsilon(H, k) \;:=\; \frac{2\epsilon|k|^{H+1}}{(H+1)!}.$$

Proof. We show that the SHC approximation error in predicting the smooth latent trend $\ell(\cdot)$ is of order $|k|^{H+1}$. The argument proceeds in four steps.

Step 1. Taylor expansion of the latent trend. Let $\ell(t)$ denote the latent smooth component of the treated outcome. By Assumption (A3), $\ell(t)$ is $(H + 1)$ times differentiable with $|\ell^{(H+1)}(t)| < \epsilon$ for all t. Expanding ℓ around T_0 using Taylor's theorem gives

$$\ell(T_0 + k) = \sum_{h=0}^{H} \frac{\ell^{(h)}(T_0)}{h!} \, k^h \;+\; R_{H+1}(k),$$

where the remainder term is

$$R_{H+1}(k) = \frac{\ell^{(H+1)}(\xi)}{(H+1)!} \, k^{H+1}, \quad \text{for some } \xi \in [T_0, T_0 + k].$$

Step 2. Bound the Taylor remainder. Since $|\ell^{(H+1)}(t)| \leq \epsilon$ for all t, the remainder is bounded by

$$|R_{H+1}(k)| \;\leq\; \frac{\epsilon}{(H+1)!} \, |k|^{H+1}.$$

Step 3. Representation of the SHC approximation. The SHC predictor $\ell^{\mathrm{SHC}}(T_0 + k)$ is constructed as a convex combination of donor segments. By design, these donor segments come from smoothed pre-treatment trajectories that share the same latent trend structure $\ell(\cdot)$. Thus, when we form ℓ^{SHC}, the lower-order terms of the Taylor expansion (up to order H) are matched exactly in expectation, and the residual discrepancy is of the same order as the Taylor remainder.

Formally, we may write

$$\ell^{\mathrm{SHC}}(T_0 + k) = \sum_{h=0}^{H} \frac{\ell^{(h)}(T_0)}{h!} \, k^h \;+\; R'_{H+1}(k),$$

where $R'_{H+1}(k)$ is another error term of order $|k|^{H+1}$.

Step 4. Combine the two remainder terms. Subtracting the SHC expansion from the true expansion gives

$$\ell(T_0 + k) - \ell^{\mathrm{SHC}}(T_0 + k) = R_{H+1}(k) - R'_{H+1}(k).$$

Applying the triangle inequality,

$$|\ell(T_0 + k) - \ell^{\mathrm{SHC}}(T_0 + k)| \;\leq\; |R_{H+1}(k)| + |R'_{H+1}(k)|.$$

Each remainder is bounded by $\frac{\epsilon}{(H+1)!} \, |k|^{H+1}$, so together we obtain

$$|\ell(T_0 + k) - \ell^{\mathrm{SHC}}(T_0 + k)| \;\leq\; \frac{2\epsilon}{(H+1)!} \, |k|^{H+1}.$$

This completes the proof. □

Remark 0.2. *Lemma 0.3 formalizes the idea that the SHC predictor inherits the smoothness of the latent trend $\ell(\cdot)$. Because SHC is built from overlapping donor segments that reflect the same underlying smooth dynamics, the approximation reproduces the first H terms of the Taylor expansion around T_0. The only discrepancy comes from the $(H+1)$-st order remainder, which is small when the latent trend is sufficiently smooth. Thus, the difference between ℓ and ℓ^{SHC} grows only polynomially in the forecast horizon k, and the error is of order $|k|^{H+1}$ rather than linear or quadratic in k. This ensures that SHC extrapolates the smooth component accurately over short to moderate horizons.*

Main Result

Proposition 0.1. *Under assumptions (A1)–(A5), the ASHC prediction error is bounded by*

$$\|\widetilde{\mathbf{y}} - \widetilde{\mathbf{Y}}\mathbf{w}^{ASHC}\| \;\leq\; \epsilon_{SHC}\left(1 + \frac{\gamma}{2\lambda + 2\varsigma\|\mathbf{C}_0\|^2}\right) + b_\epsilon(H,k) + \delta.$$

Proof. We want to establish an upper bound on the prediction error of the ASHC estimator,

$$\|\widetilde{\mathbf{y}} - \widetilde{\mathbf{Y}}\mathbf{w}^{\mathrm{ASHC}}\|.$$

The argument proceeds in four steps.

Step 1. Break down the error into two components. Let $\mathbf{w}^{\mathrm{SHC}}$ denote the SHC weights and define

$$\Delta := \mathbf{w}^{\mathrm{ASHC}} - \mathbf{w}^{\mathrm{SHC}}.$$

By the triangle inequality,

$$\|\widetilde{\mathbf{y}} - \widetilde{\mathbf{Y}}\mathbf{w}^{\mathrm{ASHC}}\| \;\leq\; \underbrace{\|\widetilde{\mathbf{y}} - \widetilde{\mathbf{Y}}\mathbf{w}^{\mathrm{SHC}}\|}_{\epsilon_{\mathrm{SHC}}} + \|\widetilde{\mathbf{Y}}\Delta\|.$$

Step 2. Bound the contribution of $\|\widetilde{\mathbf{Y}}\Delta\|$. By Lemma 0.2, we know that

$$\|\widetilde{\mathbf{Y}}\Delta\| \;\leq\; \gamma\|\Delta\|.$$

Step 3. Bound the deviation $\|\Delta\|$. By Lemma 0.1, the deviation satisfies

$$\|\Delta\| \;\leq\; \frac{\epsilon_{\mathrm{SHC}}}{2\lambda + 2\varsigma\|\mathbf{C}_0\|^2}.$$

Step 4. Account for smoothness bias and noise. From Lemma 0.3, the smoothness bias is bounded by

$$b_\epsilon(H,k) \;=\; \frac{2\epsilon|k|^{H+1}}{(H+1)!}.$$

From Assumption (A2), the noise contributes at most

$$\delta.$$

Step 5. Putting the pieces together. Substituting the bound from Step 2 into Step 1 gives:

$$\|\widetilde{\mathbf{y}} - \widetilde{\mathbf{Y}}\mathbf{w}^{\mathrm{ASHC}}\| \;\leq\; \epsilon_{\mathrm{SHC}} + \gamma\|\Delta\|.$$

Now substitute the bound for $\|\Delta\|$ from Step 3:

$$\|\widetilde{\mathbf{y}} - \widetilde{\mathbf{Y}}\mathbf{w}^{\mathrm{ASHC}}\| \;\leq\; \epsilon_{\mathrm{SHC}} + \gamma \cdot \frac{\epsilon_{\mathrm{SHC}}}{2\lambda + 2\varsigma\|\mathbf{C}_0\|^2}.$$

Next, add the smoothness bias and noise bounds from Step 4:

$$\|\widetilde{\mathbf{y}} - \widetilde{\mathbf{Y}}\mathbf{w}^{\mathrm{ASHC}}\| \;\le\; \epsilon_{\mathrm{SHC}} + \gamma \cdot \frac{\epsilon_{\mathrm{SHC}}}{2\lambda + 2\varsigma\|\mathbf{C}_0\|^2} + b_\epsilon(H,k) + \delta.$$

Step 6. Simplify the bound. Factor out ϵ_{SHC} from the first two terms:

$$\epsilon_{\mathrm{SHC}} + \gamma \cdot \frac{\epsilon_{\mathrm{SHC}}}{2\lambda + 2\varsigma\|\mathbf{C}_0\|^2} \;=\; \epsilon_{\mathrm{SHC}}\left(1 + \frac{\gamma}{2\lambda + 2\varsigma\|\mathbf{C}_0\|^2}\right).$$

Therefore,

$$\|\widetilde{\mathbf{y}} - \widetilde{\mathbf{Y}}\mathbf{w}^{\mathrm{ASHC}}\| \;\le\; \epsilon_{\mathrm{SHC}}\left(1 + \frac{\gamma}{2\lambda + 2\varsigma\|\mathbf{C}_0\|^2}\right) + b_\epsilon(H,k) + \delta.$$

This completes the proof. □

Remark 0.3. *The proof explicitly shows how each component contributes to the final bound: (1) the SHC baseline error, (2) the ASHC deviation penalty controlled by λ and ς, (3) the smoothness bias from Taylor approximation, and (4) the bounded noise. Factoring ϵ_{SHC} at the end highlights how ASHC scales the SHC error by at most a small multiplicative factor depending on the regularization strength.*

Asymptotic Results

Theorem 0.1. *Under (A1)–(A5), as $m \to \infty$ and $H \to \infty$, for fixed $k > 0$,*

$$\|\widetilde{\mathbf{y}} - \widetilde{\mathbf{Y}}w^{ASHC}\| \longrightarrow \delta.$$

If $\boldsymbol{\eta}$ consists of independent mean-zero sub-Gaussian entries, then

$$\|\widetilde{\mathbf{y}} - \widetilde{\mathbf{Y}}w^{ASHC}\| \xrightarrow{p} 0.$$

Proof. Starting from Proposition 0.1, we have

$$\|\widetilde{\mathbf{y}} - \widetilde{\mathbf{Y}}w^{\mathrm{ASHC}}\| \le \epsilon_{\mathrm{SHC}}\left(1 + \frac{\gamma}{\sqrt{2\lambda + 2\varsigma\|\mathbf{C}_0\|^2}}\right) + b_\epsilon(H,k) + \delta.$$

Step 1. Behavior of $\epsilon_{\mathbf{SHC}}$. By (A3), $\ell(t)$ is $(H+1)$-times differentiable. Gaussian kernel smoothing in SHC ensures

$$\epsilon_{\mathrm{SHC}} \to 0 \quad \text{as } m \to \infty,\ H \to \infty.$$

Chen et al. [2024]

Step 2. Behavior of ASHC penalty term. Since $\epsilon_{\mathrm{SHC}} \to 0$, the inflation factor

$$\frac{\gamma}{\sqrt{2\lambda + 2\varsigma\|\mathbf{C}_0\|^2}}\,\epsilon_{\mathrm{SHC}}$$

vanishes asymptotically.

Step 3. Behavior of smoothness bias $b_\epsilon(H,k)$. From Taylor's theorem,

$$\ell(T_0 + k) = \sum_{h=0}^{H} \frac{\ell^{(h)}(T_0)}{h!}k^h + R_{H+1}(k),$$

with

$$|R_{H+1}(k)| \le \frac{\epsilon}{(H+1)!}|k|^{H+1}.$$

Thus

$$b_\epsilon(H,k) \le \frac{2\epsilon}{(H+1)!}|k|^{H+1} \to 0 \quad \text{as } H \to \infty.$$

Step 4. Behavior of noise term. By (A2), $\|\boldsymbol{\eta}\| \le \delta$. If $\boldsymbol{\eta}$ is i.i.d. sub-Gaussian with variance proxy σ^2, then

$$\frac{\|\boldsymbol{\eta}\|}{m} \xrightarrow{p} 0.$$

Step 5. Combine bounds. Therefore,

$$\limsup_{m\to\infty} \|\widetilde{\mathbf{y}} - \widetilde{\mathbf{Y}}w^{\text{ASHC}}\| \le \delta.$$

With sub-Gaussian noise, the limit is 0 in probability. □

Corollary 0.1. *Under (A1)–(A5), the ASHC estimator is asymptotically unbiased up to δ. If $\boldsymbol{\eta}$ is i.i.d. mean-zero sub-Gaussian with variance proxy σ^2 and bandwidth $h_m \asymp m^{-1/(2H+3)}$, then as $m \to \infty$,*

$$\|\widetilde{\mathbf{y}} - \widetilde{\mathbf{Y}}w^{ASHC}\| = O_p\left(m^{-H/(2H+3)}\right).$$

For large H, this approaches the parametric rate $O_p(m^{-1/2})$.

Proof. **Part 1: Consistency.** From Theorem 0.1, the error converges to δ, or 0 in probability under sub-Gaussian noise.

Part 2: Convergence Rate. From Proposition 0.1, we have

$$\|\widetilde{\mathbf{y}} - \widetilde{\mathbf{Y}}w^{\text{ASHC}}\| \le \epsilon_{\text{SHC}}\left(1 + \frac{\gamma}{\sqrt{2\lambda + 2\varsigma\|\mathbf{C}_0\|^2}}\right) + b_\epsilon(H,k) + \delta.$$

Step 1. Bias term. From Lemma 0.3,

$$b_\epsilon(H,k) = O(h_m^H).$$

Step 2. Stochastic error term. With sub-Gaussian noise, the smoothed variance is

$$O_p\big((mh_m)^{-1/2}\big).$$

Step 3. SHC fit error. Combining,

$$\epsilon_{\text{SHC}} = O_p\big(h_m^H + (mh_m)^{-1/2}\big).$$

Step 4. Optimal bandwidth. Equating orders $h_m^H \asymp (mh_m)^{-1/2}$ yields

$$h_m \asymp m^{-1/(2H+3)}.$$

Step 5. Convergence rate. Substituting,

$$\epsilon_{\text{SHC}} = O_p\big(m^{-H/(2H+3)}\big).$$

Thus the total error is

$$\|\widetilde{\mathbf{y}} - \widetilde{\mathbf{Y}}w^{\text{ASHC}}\| = O_p\big(m^{-H/(2H+3)}\big).$$

Step 6. Near-parametric rate. As $H \to \infty$,

$$\frac{H}{2H+3} \to \tfrac{1}{2}.$$

So the rate approaches $O_p(m^{-1/2})$. □

Remark 0.4. *The deterministic noise bound δ in (A2) corresponds to $\|\boldsymbol{\eta}\| \leq \delta$. Under sub-Gaussian noise, $\delta = O(\sigma\sqrt{m})$, while kernel smoothing reduces effective noise to $O_p((mh_m)^{-1/2})$. This explains the appearance of the nonparametric rate.*

References

Yi-Ting Chen, Jui-Chung Yang, and Tzu-Ting Yang. Synthetic historical control for policy evaluation. Available at SSRN: https://ssrn.com/abstract=4995085, September 2024. URL http://dx.doi.org/10.2139/ssrn.4995085.